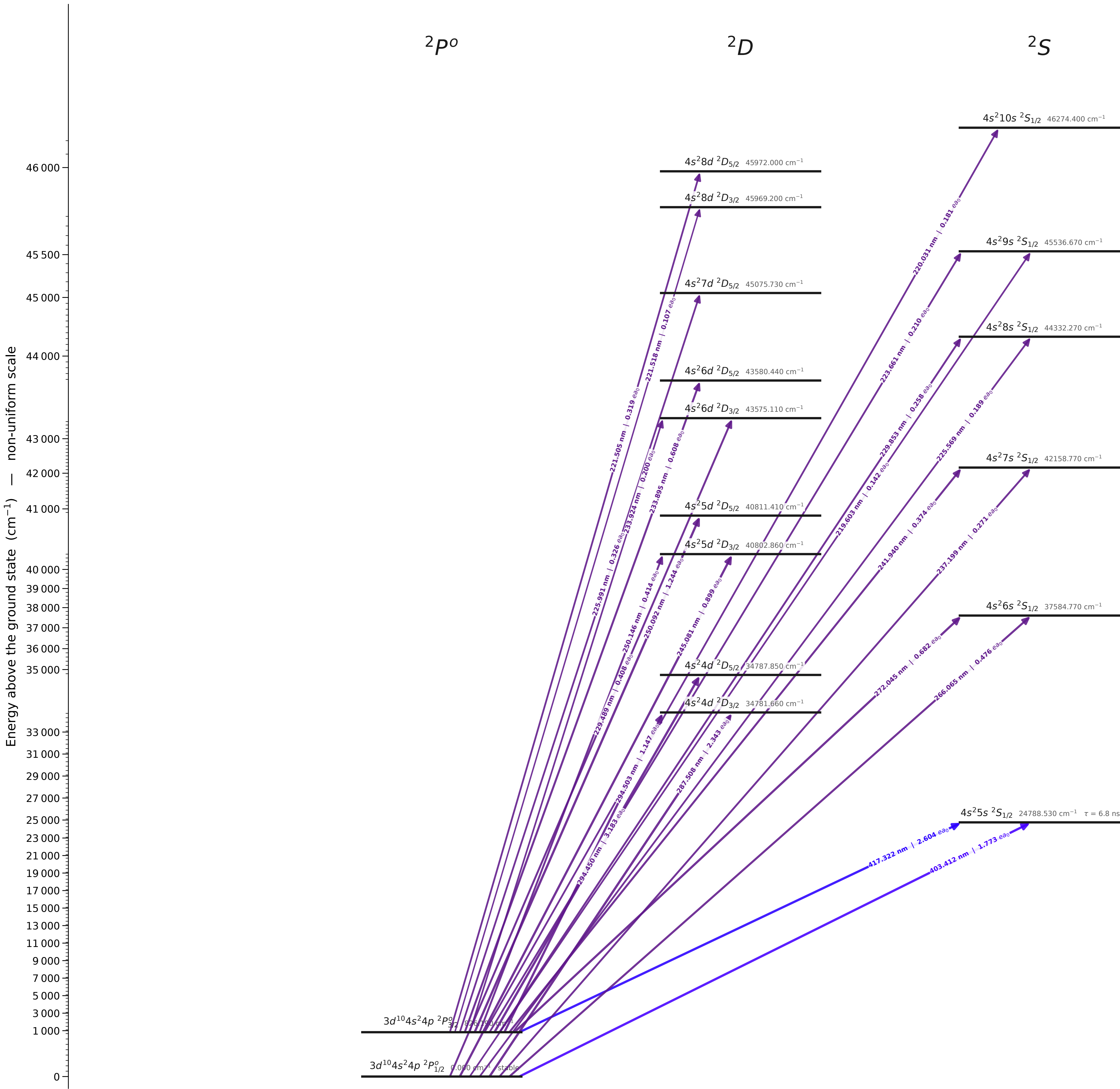


Gallium

Ga (Ga I). Strongest classified transitions below 2 μm (NIST): 17 levels, 23 lines.

Ga⁺ ionisation limit 48387.63 cm⁻¹ = 5.99930 eV (206.664 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U er J\rangle$ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$3d^{10}4s^24p^2P_{3/2}^o - 4s^24d^2D_{5/2}$	294.4497	294.3636	1018.1449	3.183	1.340e+08	-	-	294.45 nm laser-cooling
$3d^{10}4s^24p^2P_{1/2}^o - 4s^25s^2S_{1/2}$	403.4124	403.2984	743.1414	1.773	4.850e+07	23.41	33	403.3 nm resonance line
$3d^{10}4s^24p^2P_{3/2}^o - 4s^25s^2S_{1/2}$	417.3215	417.2039	718.3729	2.604	9.450e+07	23.41	64.3	417.2 nm line

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

⁶⁹Ga hyperfine structure (*I* = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
$3d^{10}4s^2 4p^2 P^o_{3/2}$	1338.9938(5)	-	F=1: -1673.74	F=2: +1004.25		
$3d^{10}4s^2 4p^2 P^o_{3/2}$	191.5(9)	63.0(20)	F=0: -639.38	F=1: -510.88	F=2: -190.88	F=3: +446.62
$4s^2 5s^2 S_{1/2}$	1069.5(15)	-	F=1: -1336.88	F=2: +802.12		
$4s^2 4d^2 D_{3/2}$	-36.3(22)	5(4)	F=0: +141.88	F=1: +100.97	F=2: +23.77	F=3: -80.53
$4s^2 4d^2 D_{5/2}$	77.3(9)	5(4)	F=1: -402.12	F=2: -251.75	F=3: -22.24	F=4: +291.20

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$3d^{10}4s^24p \rightarrow 3d^94s^24p^2D_{5/2}$	294.450	71-69	115(7)
$3d^{10}4s^24p \rightarrow 3d^94s^24p^2D_{3/2}$	294.503	71-69	114(8)
$3d^{10}4s^24p \rightarrow 3d^94s^25s^2S_{1/2}$	403.412	69-71	33(4)
$3d^{10}4s^24p \rightarrow 3d^94s^25s^2S_{1/2}$	417.322	69-71	40(4)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (wh